

# Solomonoff Induction

David Quarel

Australian National University

Leon Lang

University of Amsterdam

September 24, 2026

## Prerequisites

- Comfort with discrete probability: conditional distributions, the chain rule, expectations.
- Kullback–Leibler divergence and basic information theory (helpful, developed as needed).

### What you'll learn

- Define the Bayesian mixture over a class of environments and prove it is a proper predictor whose posterior updates multiplicatively.
- Prove the cumulative prediction-error bound and specialize it to the Solomonoff prior to recover the Occam bound.
- Show the mixture makes boundedly many mistakes and is Pareto-optimal under KL and squared loss.
- Bound the misspecified case when the truth lies outside the model class.

Much of this material is drawn from [HQC24, Chapter 3] and the earlier [Hut05]; both books provide fuller explanations, additional context, and proofs that we gloss over here.

**Difficulty ratings.** Each problem is tagged with a rating in square brackets following Knuth's exercise rating scheme [Knu73]: roughly, [00] trivial, [10] 15-minute pencil-and-paper, [20] 1–2 hours, [30] several hours to a day, [40] a significant research result. Intermediate values are possible.

# Overview

Solomonoff induction is the *prediction* half of **Universal AI**: what would an *optimally intelligent* agent do if it only had to predict, given unlimited compute and the weakest possible assumptions? The answer is a single **Bayesian mixture**  $\xi$  over a countable class  $\mathcal{M}$  of candidate environments, weighted by a prior  $w_\nu$ . It (eventually) predicts as well as the true environment  $\mu$ , and under the **universal** choice —  $\mathcal{M}$  = all computable environments, prior  $w_\nu = 2^{-K(\nu)}$  with  $K$  the Kolmogorov complexity — the assumption “ $\mu \in \mathcal{M}$ ” becomes “the universe is computable,” and Occam’s razor drops out of the mathematics.

This module is a worksheet: you build the theory yourself, one problem at a time. The sequel module, [AIXI](#), lifts the same mixture to sequential decision-making (learning to *act*).

## Goal and Roadmap

**A recipe for prediction:** Solomonoff induction is an attempt to mathematically formalize the *problem of induction* from philosophy: How to make predictions about the future based on past observations?

We formalize this as sequence prediction: There is some true environment  $\mu$  that generates a sequence  $x_1, x_2, \dots$  of (binary) symbols. Each next symbol  $x_t \sim \mu(\cdot \mid x_{<t})$  is sampled from  $\mu$  conditioned on the past  $x_{<t}$ . A predictor  $P$  takes the history  $x_{<t}$  and gives a distribution  $P(\cdot \mid x_{<t})$  over the next symbol  $x_t$ .

We measure the quality of a predictor  $P$  by the  $\mu$ -expected squared prediction error, summed over every timestep:

$$S_\infty^\mu := \sum_{t=1}^{\infty} \sum_{x_{<t} \in \mathbb{B}^*} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} (P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}))^2. \quad (1)$$

How to construct a predictor  $P$  such that  $S_\infty^\mu$  is small (or at least finite)? Bayesian inference to the rescue: we choose as our predictor a *Bayesian mixture*  $\xi$  over a countable class  $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$  of candidate environments, weighted by prior beliefs  $w_\nu$  (formalized in Definition 1.2). The main results we build up to are:

- **Cumulative bound** (Section 4): Assuming  $\mu \in \mathcal{M}$ ,  $S_\infty^\mu \leq -\ln w_\mu$ . The higher the prior on  $\mu$ , the lower the prediction error.
- **Explicit bound** (Exercise 4.2): specialize to the Solomonoff prior  $w_\nu = 2^{-K(\nu)}$  to get  $S_\infty^\mu \leq K(\mu) \ln 2$ , where  $K$  is the *Kolmogorov complexity*.

- **Pareto optimality** (Exercises 6.2 and 8.2): no other predictor weakly dominates  $\xi$  on every  $\nu \in \mathcal{M}$ , for either KL or squared loss.
- **Misspecified version** (Section 7): If  $\mu \notin \mathcal{M}$ , the cumulative bound becomes  $-\ln w_{\hat{\mu}} + D_n(\mu \parallel \hat{\mu})$ : the constant complexity term plus an approximation term  $D_n(\mu \parallel \hat{\mu}) = \sum_{t=1}^n d_t(\mu \parallel \hat{\mu})$  that in general grows linearly in  $n$ , so  $S_{\infty}^{\mu}$  diverges. Here  $\hat{\mu} \in \mathcal{M}$  is the “closest” environment to  $\mu$ .

## Notation

For more background, see [the corresponding post on Solomonoff induction](#).

### Note

#### Symbols used throughout.

- $\mathbb{B} = \{0, 1\}$ : the binary alphabet;  $\mathbb{B}^*$  all finite binary strings;  $\mathbb{B}^n$  length- $n$  strings.
- $\epsilon$ : the empty string.
- $xy$ : string concatenation. if  $x := x_{1:n}$  and  $y := y_{1:m}$  then  $xy = x_1 \dots x_n y_1 \dots y_m$ .
- $x_{i:j} := x_i x_{i+1} \dots x_j$ .
- $x_{<t} := x_1 x_2 \dots x_{t-1}$ .
- $\nu, \rho$ : generic environments / predictors.  $\mu$ : the true (unknown) environment generating the data.  $\xi$ : a Bayesian mixture of environments.
- $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ : a countable class of candidate environments.
- $w_{\nu} > 0$  with  $\sum_{\nu \in \mathcal{M}} w_{\nu} = 1$ : the prior over  $\mathcal{M}$ .  $w(\nu \mid x_{<t})$ : the posterior after observing  $x_{<t}$ .
- $\Delta\mathcal{X}$ : the set of probability distributions over a finite set  $\mathcal{X}$ .

## 1. The mixture is a predictor

**Definition 1.1** (Environment). An **environment**  $\nu$  assigns to each history  $x_{<t} \in \mathbb{B}^*$  a predictive distribution  $\nu(\cdot \mid x_{<t}) \in \Delta\mathbb{B}$  over the next symbol. The conditional joint probability of a string  $x \in \mathbb{B}^*$ , conditioned on  $y \in \mathbb{B}^*$ , is defined

as

$$\nu(x_{1:n}|y) := \prod_{t=1}^n \nu(x_t | yx_{<t}), \quad \nu(\epsilon) := 1.$$

The unconditional joint is defined as  $\nu(x_{1:n}) := \nu(x_{1:n} | \epsilon)$ . The true (unknown) environment generating the data is denoted  $\mu$ .

### Remark

One can show that  $\nu(x_{1:t}) = \nu(x_{<t}) \cdot \nu(x_t | x_{<t})$  and  $\nu(x) = \nu(x0) + \nu(x1)$  for all  $x \in \mathbb{B}^*$ .

**Definition 1.2** (Bayesian mixture  $\xi$ ). Let  $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$  be a countable class of environments with prior weights  $w_\nu > 0$  satisfying  $\sum_{\nu \in \mathcal{M}} w_\nu = 1$ . The **Bayesian mixture** is defined as a prior-weighted mixture over all environments  $\nu$  in  $\mathcal{M}$ :

$$\xi(x_{1:t}) := \sum_{\nu \in \mathcal{M}} w_\nu \nu(x_{1:t}). \quad (2)$$

Its one-step predictive distribution is

$$\xi(x_t | x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu | x_{<t}) \nu(x_t | x_{<t}),$$

with posterior weights

$$w(\nu | x_{<t}) := w_\nu \frac{\nu(x_{<t})}{\xi(x_{<t})} \quad w(\nu | \epsilon) := w_\nu.$$

Here  $w(\nu | x_{<t})$  is the *posterior* belief in  $\nu$  after observing  $x_{<t}$ .

The mixture  $\xi$  is defined as a sum of *joint* probabilities. To use it as a predictor we need its one-step conditional  $\xi(x_t | x_{<t})$ , and we need to know it is a genuine probability distribution (so that, later, Pinsker's inequality applies to it).

### Exercise 1.1 (Generalized chain rule). [05]

The two-term chain rule baked into Definition 1.1 extends to arbitrary contiguous blocks. Show that for any  $1 \leq p \leq q \leq r$  and any history  $x_{<p}$  with  $\nu(x_{<p}) > 0$ ,

$$\nu(x_{p:r} | x_{<p}) = \nu(x_{p:q} | x_{<p}) \cdot \nu(x_{q+1:r} | x_{<p} x_{p:q}).$$

### Exercise 1.2 (Mixture is a predictor). [10]

Starting from  $\xi(x_{1:t}) = \sum_{\nu \in \mathcal{M}} w_\nu \nu(x_{1:t})$ , show that

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \nu(x_t \mid x_{<t}), \quad w(\nu \mid x_{<t}) := w_\nu \frac{\nu(x_{<t})}{\xi(x_{<t})},$$

and conclude that  $\sum_{x_t \in \mathbb{B}} \xi(x_t \mid x_{<t}) = 1$ , i.e.  $\xi(\cdot \mid x_{<t})$  is a probability distribution over  $\mathbb{B}$ .

**Hint:** Write  $\xi(x_t \mid x_{<t}) = \xi(x_{1:t})/\xi(x_{<t})$ , expand the numerator, and use the chain rule  $\nu(x_{1:t}) = \nu(x_{<t}) \nu(x_t \mid x_{<t})$  (Definition 1.1). For the last part, note that the posterior weights sum to 1.

### Exercise 1.3 (Posterior update). [10]

Show that the posterior weight  $w(\cdot \mid \cdot)$  updates *multiplicatively* as new symbols arrive. For every  $x_{1:t} \in \mathbb{B}^*$  with  $\xi(x_{1:t}) > 0$ ,

$$w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}.$$

That is: the new posterior equals the old posterior, scaled by the **likelihood ratio** of how well  $\nu$  predicted the just-observed symbol relative to the mixture's own prediction.

### Exercise 1.4 (Multi-step posterior linearity). [10]

Exercise 1.2 showed that  $\xi$ 's one-step prediction  $\xi(x_t \mid x_{<t})$  is the posterior-weighted average of the per-model one-step predictions. The same identity extends to predictions over a *whole future segment*  $x_{t:m}$ : for any  $t \leq m$ ,

$$\xi(x_{t:m} \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \nu(x_{t:m} \mid x_{<t}).$$

That is: the *same* posterior  $w(\nu \mid x_{<t})$  governs  $\xi$ 's predictions for arbitrarily many steps into the future, not just the next one. Show this.

**Hint:** Write  $\xi(x_{t:m} \mid x_{<t}) = \xi(x_{1:m})/\xi(x_{<t})$ , expand the numerator using the mixture definition, and apply the generalized chain rule (Exercise 1.1 with  $p = 1$ ,  $q = t - 1$ ,  $r = m$ ) to factor each  $\nu(x_{1:m}) = \nu(x_{<t}) \nu(x_{t:m} \mid x_{<t})$ .

## 2. KL divergence

The KL divergence between two joint distributions over a length- $n$  history splits, by an inductive application of the chain rule for probabilities, into a sum of per-step conditional KLs. This telescoping identity lets us trade a single joint KL over an entire history for a sum of per-step KLs (and vice versa): exactly the bridge we will need in Section 4.

We introduce two pieces of notation that will be used throughout the rest of the worksheet:

**Definition 2.1** (Cumulative and per-step KL). For environments / predictors  $\nu, \rho$ , horizon  $n \geq 1$ , and  $1 \leq t \leq n$ , the **cumulative joint KL** and the **expected per-step KL at step  $t$**  are

$$D_n(\nu \parallel \rho) := \sum_{x_{1:n} \in \mathbb{B}^n} \nu(x_{1:n}) \ln \frac{\nu(x_{1:n})}{\rho(x_{1:n})},$$

$$d_t(\nu \parallel \rho) := \sum_{x_{<t} \in \mathbb{B}^{t-1}} \nu(x_{<t}) \sum_{x_t \in \mathbb{B}} \nu(x_t \mid x_{<t}) \ln \frac{\nu(x_t \mid x_{<t})}{\rho(x_t \mid x_{<t})},$$

with the conventions  $0 \ln \frac{0}{q} := 0$  for  $q \geq 0$  and  $p \ln \frac{p}{0} := \infty$  for  $p > 0$ . We write  $D_\infty(\nu \parallel \rho) := \lim_{t \rightarrow \infty} D_t(\nu \parallel \rho)$  and  $D_\infty^\mu := D_\infty(\mu \parallel \xi)$ .

**Theorem 2.2** (KL divergence is non-negative). *For any environments  $\nu, \rho$  and any horizon  $n \geq 1$  and step  $1 \leq t \leq n$ :*

- (a)  $D_n(\nu \parallel \rho) \geq 0$ , with equality iff  $\nu(x_{1:n}) = \rho(x_{1:n})$  for every  $x_{1:n} \in \mathbb{B}^n$  with  $\nu(x_{1:n}) > 0$ .
- (b)  $d_t(\nu \parallel \rho) \geq 0$ , with equality iff  $\nu(x_t \mid x_{<t}) = \rho(x_t \mid x_{<t})$  for every  $x_{<t}$  with  $\nu(x_{<t}) > 0$  and every  $x_t \in \mathbb{B}$ .

*Proof in Appendix B.*

**Exercise 2.1 (Telescoping KL).** [10]

Show by induction on  $n$  that for all environments  $\nu, \rho$ ,

$$D_n(\nu \parallel \rho) = \sum_{t=1}^n d_t(\nu \parallel \rho),$$

where on the right-hand side, the  $t = 1$  summand uses the conventions  $\nu(x_1 \mid x_{<1}) := \nu(x_1)$ ,  $\rho(x_1 \mid x_{<1}) := \rho(x_1)$ .

**Hint:** For the induction step, factor  $\nu(x_{1:n}) = \nu(x_{<n}) \nu(x_n \mid x_{<n})$  (and likewise for  $\rho$ ) inside the log; the log splits additively into two pieces matching  $D_{n-1}(\nu \parallel \rho)$  and  $d_n(\nu \parallel \rho)$ .

### 3. Mixture dominance

The mixture  $\xi$  never assigns much less probability than any single environment weighted by its prior. This single inequality is the engine behind the cumulative bound of Section 4 below: dividing through gives  $\mu(x)/\xi(x) \leq 1/w_\mu$ , which is then fed into Pinsker and the chain rule in the proof of the main cumulative bound.

**Exercise 3.1 (Mixture dominance). [05]**

Show that for every  $\nu \in \mathcal{M}$  and every  $x \in \mathbb{B}^*$ ,

$$\xi(x) \geq w_\nu \cdot \nu(x).$$

**Remark (Solomonoff specialization)**

Under the Solomonoff prior,  $w_\nu = 2^{-K(\nu)}$ , so the bound becomes

$$\xi_U(x) \geq 2^{-K(\nu)} \cdot \nu(x) \quad \text{for every computable } \nu.$$

This is what makes  $\xi_U$  *universal*: a single predictor dominates the entire computable model class up to a factor that depends only on the model's description length.

### 4. Cumulative prediction error bound (main result)

The next ingredient relates squared prediction error to KL divergence [CT06, Lemma 11.6.1]. We require the following inequality:

**Theorem 4.1** (Pinsker's inequality). *Let  $P$  and  $Q$  be probability distributions over  $\mathbb{B} = \{0, 1\}$ , i.e.,  $p_0, p_1, q_0, q_1 \geq 0$  with  $p_0 + p_1 = 1$  and  $q_0 + q_1 = 1$ . Write  $p_x = P(x)$  and  $q_x = Q(x)$ . Then*

$$\sum_{x \in \mathbb{B}} (q_x - p_x)^2 \leq \sum_{x \in \mathbb{B}} p_x \ln \frac{p_x}{q_x}. \quad (\star)$$

The proof of this inequality is mostly tedious algebra but is included in Appendix A.

**Definition 4.2** (Cumulative expected and per-step squared prediction error). For environments / predictors  $\nu, \rho$  and horizon  $n \geq 1$ , the **cumulative expected squared error** and the **per-step squared error at step  $t$**  are

$$S_n(\nu \parallel \rho) := \sum_{t=1}^n s_t(\nu \parallel \rho)$$

$$s_t(\nu \parallel \rho) := \sum_{x_{<t} \in \mathbb{B}^{t-1}} \nu(x_{<t}) \sum_{x_t \in \mathbb{B}} (\nu(x_t \mid x_{<t}) - \rho(x_t \mid x_{<t}))^2$$

Write  $S_\infty(\nu \parallel \rho) := \lim_{n \rightarrow \infty} S_n(\nu \parallel \rho)$  and  $S_\infty^\mu := S_\infty(\mu \parallel \xi)$ .

**Exercise 4.1 (Cumulative prediction bound).** [15]

Show that

$$S_\infty^\mu \leq -\ln w_\mu.$$

**Hint:** You need Theorem 4.1, Exercise 2.1 and Exercise 3.1.

**Exercise 4.2 (Explicit complexity bound).** [05]

Specialize Section 4 to the **Solomonoff prior**  $w_\nu = 2^{-K(\nu)}$  (where  $K(\nu)$  is the length of the shortest program computing  $\nu$ ) to show

$$S_\infty^\mu \leq K(\mu) \ln 2.$$

**Exercise 4.3 (Per-step error).** [05]

Hence prove that the per-step expected prediction error

$$s_t := \sum_{x_{<t} \in \mathbb{B}^{t-1}} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} (\xi(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}))^2$$

converges to zero as  $t \rightarrow \infty$ .

## 5. Bounded number of prediction mistakes

So far we have bounded *squared error*. For a deterministic environment (i.e. a single infinite binary sequence  $x_1^*, x_2^*, \dots$  in  $\mathcal{M}$ ), one can ask the sharper 0/1-loss question: how often does the mixture predict the *wrong bit*? The cumulative bound is strong enough to give an answer.

Let  $\mu \in \mathcal{M}$  be a deterministic environment, so  $\mu(x_t \mid x_{<t}) \in \{0, 1\}$  on every history;

write  $x_t^*$  for the unique bit on which  $\mu(\cdot \mid x_{<t}^*)$  puts all its mass. Define the **threshold predictor** associated with  $\xi$ : at each step, predict the more-likely bit,

$$\hat{x}_t := \arg \max_{x \in \mathbb{B}} \xi(x \mid x_{<t}) \quad (\text{break ties arbitrarily}).$$

We say the mixture makes a **mistake** at time  $t$  if  $\hat{x}_t \neq x_t^*$ .

**Exercise 5.1 (Bounded prediction mistakes).** [15]

Show that the number of mistakes made by the threshold predictor on the  $\mu$ -trajectory is at most

$$\#\{t \geq 1 : \hat{x}_t \neq x_t^*\} \leq -2 \ln w_\mu.$$

**Hint:** At a mistake step,  $\xi(x_t^* \mid x_{<t}) \leq \frac{1}{2}$  (why?). The per-step squared error along the deterministic  $\mu$ -trajectory simplifies dramatically; bound it from below, then sum.

**Remark (Solomonoff specialization)**

Under the Solomonoff prior  $w_\mu = 2^{-K(\mu)}$ , so  $-\ln w_\mu = K(\mu) \ln 2$  and the bound becomes

$$\#\{t \geq 1 : \hat{x}_t \neq x_t^*\} \leq 2K(\mu) \ln 2.$$

With  $\ln 2 \approx 0.693$  the constant is just under 1.4, so if the true environment  $\mu$  is computable, and can be described by a program at most  $k$  bits long, then the predictor  $\xi$  will make at worst  $\approx 1.4k$  mistakes over predicting the entire sequence  $x_{1:\infty}^*$ . Note that the bound is purely existence-style: it does not say *when* the mistakes happen. They could all occur at the start, or be arbitrarily far into the future.

## 6. Pareto optimality under KL loss

In Section 6 and Section 8 we establish that the Bayesian mixture  $\xi$  is *uniquely Pareto-optimal* under both KL and squared prediction loss: no other predictor can do at least as well as  $\xi$  on every environment  $\nu \in \mathcal{M}$  without coinciding with  $\xi$  entirely.

**Definition 6.1** (Pareto domination). For some measure of **loss**  $\mathcal{F}(\nu, \rho)$  that takes an environment  $\nu$  and a predictor  $\rho$ , a predictor  $\rho$  **weakly Pareto-dominates**  $\xi$  with respect to loss  $\mathcal{F}$  and class  $\mathcal{M}$  if

$$\mathcal{F}(\nu, \rho) \leq \mathcal{F}(\nu, \xi) \quad \text{for every } \nu \in \mathcal{M}.$$

$\xi$  is **Pareto-optimal** (w.r.t.  $\mathcal{F}$ ) if the only predictor that weakly dominates it

is  $\xi$  itself.

We specialize  $\mathcal{F}$  to  $D_n(\nu \parallel \rho)$  in Section 6 and to  $S_n(\nu \parallel \rho)$  in Section 8.

**Exercise 6.1 (KL Pythagorean identity).** [10]

For *any* predictor  $\rho$  with  $\rho(x_{1:n}) > 0$  on every  $x_{1:n} \in \mathbb{B}^n$ , show that

$$D_n(\xi \parallel \rho) + \sum_{\nu \in \mathcal{M}} w_\nu D_n(\nu \parallel \xi) = \sum_{\nu \in \mathcal{M}} w_\nu D_n(\nu \parallel \rho).$$

**Hint:** Take the difference between the two summation terms.

**Exercise 6.2 (KL Pareto-optimality).** [05]

Show that  $\xi$  is Pareto-optimal under  $D_n(\nu \parallel \rho)$  in the sense of Definition 6.1: if  $\rho$  is any predictor with

$$D_n(\nu \parallel \rho) \leq D_n(\nu \parallel \xi) \quad \text{for every } \nu \in \mathcal{M},$$

then  $\rho = \xi$ .

**Hint:** Multiply the assumed inequality by  $w_\nu \geq 0$  and sum over  $\nu \in \mathcal{M}$ ; then apply the KL Pythagorean identity to recognize the extra non-negative term, which must vanish.

## 7. Misspecified models: when $\mu \notin \mathcal{M}$

The cumulative bound Section 4 assumed the true environment  $\mu$  lives in the model class  $\mathcal{M}$ . What happens when it does not? We now show the bound *degrades gracefully*: it splits cleanly into a complexity term (“cost of not knowing which model is best”), exactly as before, plus a linear-in-time approximation term (“cost of no model being right”). See [Hut05, §3.2.8] for the original treatment.

Throughout this section, we no longer assume  $\mu \in \mathcal{M}$ . We will state the bound in terms of an arbitrary  $\hat{\mu} \in \mathcal{M}$ , so taking the infimum over  $\hat{\mu}$  on the right-hand side gives the tightest version by using the “closest” approximation of  $\hat{\mu} \in \mathcal{M}$  to  $\mu$ .

**Exercise 7.1 (Misspecified KL bound).** [10]

Show that for all  $\hat{\mu} \in \mathcal{M}$ ,

$$D_n(\mu \parallel \xi) \leq -\ln w_{\hat{\mu}} + D_n(\mu \parallel \hat{\mu}).$$

**Hint:** Use Exercise 3.1.

### Remark (Reading the two terms)

The right-hand side splits into two qualitatively different terms:

- $-\ln w_{\hat{\mu}}$  is *constant in  $n$* : under the Solomonoff prior, this is  $K(\hat{\mu}) \ln 2$ . The familiar “complexity” cost of search.
- The approximation term  $D_n(\mu \parallel \hat{\mu}) = \sum_{t=1}^n d_t(\mu \parallel \hat{\mu})$  (by Exercise 2.1) is a sum of  $n$  per-step KLs; in general it grows with  $n$ . It vanishes identically iff  $\hat{\mu}$  matches  $\mu$  on every  $\mu$ -reachable history, in particular when  $\mu \in \mathcal{M}$  and we pick  $\hat{\mu} = \mu$ , recovering Section 4.

Combining with Pinsker (Theorem 4.1) gives the corresponding bound on  $S_{\infty}^{\mu}$ , which is infinite in general but inherits the rate of  $\hat{\mu}$ .

Details on how to define the best choice of  $\hat{\mu}$  are in Appendix D.

## 8. Pareto optimality under squared loss

The squared specialization  $\mathcal{F}(\nu, \rho) = S_n(\nu \parallel \rho)$  (Definition 4.2) lives one level down from KL: at the conditional distributions  $\nu(\cdot \mid x_{<t})$ . The natural Pythagorean decomposition at a fixed history uses *posterior* weights  $w(\nu \mid x_{<t})$ : those are the weights that make  $\xi(x_t \mid x_{<t})$  the mean of the  $\nu(x_t \mid x_{<t})$ ’s (via the posterior-predictive form, Definition 1.2). The Pareto-optimality aggregation, however, uses prior weights  $w_{\nu}$ . Bridging the two takes one extra Bayes-rule step.

### Exercise 8.1 (Squared Pythagorean identity). [10]

Fix any  $t \in \{1, \dots, n\}$  and any history  $x_{<t} \in \mathbb{B}^{t-1}$ : *the history is arbitrary, not sampled from any environment*. For any  $x_t \in \mathbb{B}$  and any predictor  $\rho$ , show

$$\begin{aligned} & (\xi(x_t \mid x_{<t}) - \rho(x_t \mid x_{<t}))^2 + \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) (\nu(x_t \mid x_{<t}) - \xi(x_t \mid x_{<t}))^2 \\ &= \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) (\nu(x_t \mid x_{<t}) - \rho(x_t \mid x_{<t}))^2. \end{aligned}$$

**Hint:** Add and subtract  $\xi(x_t \mid x_{<t})$  inside the squared term, then expand. Use  $\sum_{\nu} w(\nu \mid x_{<t}) = 1$  and the posterior-predictive form of  $\xi$  (Definition 1.2),  $\sum_{\nu} w(\nu \mid x_{<t}) \nu(x_t \mid x_{<t}) = \xi(x_t \mid x_{<t})$ , to kill the cross-term. The weighting must be *posterior* weights  $w(\nu \mid x_{<t})$  (not prior weights  $w_{\nu}$ ), because  $\xi(x_t \mid x_{<t})$  is the posterior-weighted mean of the  $\nu(x_t \mid x_{<t})$ ’s, not the prior one.

**Exercise 8.2 (Squared Pareto-optimality). [05]**

Show that  $\xi$  is Pareto-optimal under  $S_n(\nu \parallel \rho)$  in the sense of Definition 6.1: if  $\rho$  is any predictor with

$$S_n(\nu \parallel \rho) \leq S_n(\nu \parallel \xi) \quad \text{for every } \nu \in \mathcal{M},$$

then  $\rho(x_t \mid x_{<t}) = \xi(x_t \mid x_{<t})$  at every history  $x_{<t} \in \mathbb{B}^{t-1}$  with  $\xi(x_{<t}) > 0$  (equivalently,  $\xi$ -almost surely) and every  $x_t \in \mathbb{B}$ . Histories that the mixture never reaches ( $\xi(x_{<t}) = 0$ , i.e., reached by no  $\nu \in \mathcal{M}$  either) are invisible to the loss and are not pinned down.

**Hint:** Multiply the assumed inequality by  $w_\nu \geq 0$  and sum over  $\nu$ ; then for each fixed history  $x_{<t}$ , multiply the per-history Pythagorean identity from above by  $\xi(x_{<t})$  and use the Bayes identity  $w(\nu \mid x_{<t}) \xi(x_{<t}) = w_\nu \nu(x_{<t})$  to bridge from posterior weights (inside the identity) to prior weights (in the aggregated Pareto inequality).

## A. Proof of Pinsker's inequality

We first prove the following lemma:

**Lemma A.1** (Pinsker's binary inequality). *For  $0 \leq p \leq 1$  and  $0 < q < 1$  we have*

$$2(p - q)^2 \leq p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q}.$$

Define  $f(x) := p \ln x + (1 - p) \ln(1 - x)$ , so that

$$p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q} = f(p) - f(q) = \int_q^p f'(x) dx = \int_q^p \frac{p - x}{x(1 - x)} dx,$$

using  $f'(x) = \frac{p}{x} - \frac{1-p}{1-x} = \frac{p-x}{x(1-x)}$ . Recall also that  $x(1-x) \leq \frac{1}{4}$  on  $[0, 1]$ , so  $\frac{1}{x(1-x)} \geq 4$ .

**Case  $q \leq p$ .** For every  $x \in [q, p]$  we have  $x \leq p$ , hence  $p - x \geq 0$ .

$$\frac{p - x}{x(1 - x)} \geq 4(p - x) \quad \text{for } x \in [q, p].$$

Integrating over  $[q, p]$ ,

$$\int_q^p \frac{p - x}{x(1 - x)} dx \geq 4 \int_q^p (p - x) dx = 2(p - q)^2.$$

**Case  $q \geq p$ .** For every  $x \in [p, q]$  we have  $x \geq p$ , hence  $x - p \geq 0$ .

$$\frac{x - p}{x(1 - x)} \geq 4(x - p) \quad \text{for } x \in [p, q].$$

Integrating over  $[p, q]$ ,

$$\begin{aligned} \int_q^p \frac{p - x}{x(1 - x)} dx &= \int_p^q \frac{x - p}{x(1 - x)} dx \\ &\geq 4 \int_p^q (x - p) dx = 4 \int_q^p (p - x) dx = 2(p - q)^2 \end{aligned}$$

*Proof of Theorem 4.1.* We now reduce Theorem 4.1 to the binary inequality Lemma A.1. Recall  $p_0 + p_1 = 1$  and  $q_0 + q_1 = 1$ . We handle the boundary case  $q_x = 0$  first, then reduce the interior case directly to Lemma A.1.

**Case 1:  $q_x = 0$  for some  $x \in \mathbb{B}$ .** If  $p_x > 0$ , then the right-hand side of  $(\star)$  contains  $p_x \ln \frac{p_x}{0} = +\infty$ , so  $(\star)$  holds trivially. If instead  $p_x = 0$ , that atom contributes 0 to both sides (using the convention  $0 \ln \frac{0}{0} = 0$ ). The other atom  $x'$  then satisfies  $p_{x'} = 1$  and  $q_{x'} = 1 - q_x = 1$ , so both sides vanish and  $(\star)$  holds.

**Case 2:**  $q_0, q_1 > 0$ . Since  $q_1 = 1 - q_0$  and  $p_1 = 1 - p_0$ , the two sides of  $(\star)$  become

$$\sum_{x \in \mathbb{B}} (q_x - p_x)^2 = (q_0 - p_0)^2 + ((1 - q_0) - (1 - p_0))^2 = 2(p_0 - q_0)^2,$$

$$\sum_{x \in \mathbb{B}} p_x \ln \frac{p_x}{q_x} = p_0 \ln \frac{p_0}{q_0} + (1 - p_0) \ln \frac{1 - p_0}{1 - q_0}.$$

So  $(\star)$  is exactly  $2(p_0 - q_0)^2 \leq p_0 \ln \frac{p_0}{q_0} + (1 - p_0) \ln \frac{1 - p_0}{1 - q_0}$ , which is Lemma A.1 with  $p = p_0$  and  $q = q_0$  (here  $0 < q_0 < 1$ , since  $q_0, q_1 > 0$ ).  $\square$

## B. Proof of KL non-negativity

We prove Theorem 2.2. The core is the elementary inequality

$$\ln x \leq x - 1 \quad \text{for all } x > 0, \quad \text{equality iff } x = 1.$$

(Let  $f(x) := (x - 1) - \ln x$ ; then  $f'(x) = 1 - 1/x$ , vanishing only at  $x = 1$ , with  $f''(x) = 1/x^2 > 0$ , so  $f$  is strictly convex and attains its unique minimum  $f(1) = 0$  at  $x = 1$ .)

**Lemma B.1** (Gibbs' inequality). *Let  $P, Q$  be probability distributions over a finite set  $\mathcal{X}$ , with the conventions  $0 \ln \frac{0}{q} := 0$  and  $p \ln \frac{p}{0} := \infty$ . Then*

$$\sum_{x \in \mathcal{X}} P(x) \ln \frac{P(x)}{Q(x)} \geq 0,$$

*with equality iff  $P(x) = Q(x)$  for every  $x$  with  $P(x) > 0$ .*

*Proof.* If  $Q(x) = 0$  for some  $x$  with  $P(x) > 0$ , the LHS is  $+\infty$  and the inequality is strict. Otherwise restrict the sum to  $\{x : P(x) > 0\}$  (terms with  $P(x) = 0$  contribute 0 by convention), where every ratio  $Q(x)/P(x)$  is in  $(0, \infty)$ . Apply  $\ln y \leq y - 1$  to  $y = Q(x)/P(x)$ :

$$\begin{aligned} - \sum_x P(x) \ln \frac{P(x)}{Q(x)} &= \sum_x P(x) \ln \frac{Q(x)}{P(x)} \leq \sum_x P(x) \left( \frac{Q(x)}{P(x)} - 1 \right) \\ &= \sum_x Q(x) - \sum_x P(x) \leq 0, \end{aligned}$$

where the last step uses  $\sum_x Q(x) \leq 1 = \sum_x P(x)$  (the sum over  $\{x : P(x) > 0\}$  may miss some  $Q$ -mass). Negating gives the claim. Equality in  $\ln y \leq y - 1$  requires  $y = 1$  for every  $x$  with  $P(x) > 0$ , i.e.  $Q(x) = P(x)$  there; equality in  $\sum Q \leq \sum P$  then forces  $Q$  to have no mass outside  $\text{supp } P$ . Both together give  $P = Q$  on all of  $\mathcal{X}$ .  $\square$

*Proof of Theorem 2.2. Part (a).* Apply Lemma B.1 with  $\mathcal{X} = \mathbb{B}^n$ ,  $P(x_{1:n}) = \nu(x_{1:n})$ ,  $Q(x_{1:n}) = \rho(x_{1:n})$ . Both are probability distributions on  $\mathbb{B}^n$  since  $\nu$  and  $\rho$  are environments. The lemma gives  $D_n(\nu \parallel \rho) \geq 0$  with the stated equality condition.

*Part (b).* For each fixed history  $x_{<t}$  with  $\nu(x_{<t}) > 0$ , apply Lemma B.1 with  $\mathcal{X} = \mathbb{B}$ ,  $P = \nu(\cdot \mid x_{<t})$ ,  $Q = \rho(\cdot \mid x_{<t})$  (both are probability distributions on  $\mathbb{B}$ ):

$$\sum_{x_t \in \mathbb{B}} \nu(x_t \mid x_{<t}) \ln \frac{\nu(x_t \mid x_{<t})}{\rho(x_t \mid x_{<t})} \geq 0,$$

with equality iff  $\nu(\cdot \mid x_{<t}) = \rho(\cdot \mid x_{<t})$ . Multiply by  $\nu(x_{<t}) \geq 0$  and sum over  $x_{<t}$ : this gives  $d_t(\nu \parallel \rho) \geq 0$ , as a sum of non-negative terms. Equality forces every  $x_{<t}$  with  $\nu(x_{<t}) > 0$  to contribute a vanishing inner sum, which is exactly the stated condition.  $\square$

## C. Model Class $\mathcal{M}$

All results in this sheet hold for any countable class  $\mathcal{M}$  and prior  $w_\nu$ . The canonical **Solomonoff** choice takes  $\mathcal{M}$  to be the class of all computable environments and the **Solomonoff prior**

$$w_\nu := 2^{-K(\nu)},$$

where  $K(\nu)$  is the *Kolmogorov complexity* of  $\nu$ : the length of the shortest program that computes  $\nu$ . The Bayesian mixture  $\xi$  under this particular prior is the *universal mixture*, written  $\xi_U$ . With this class the assumption  $\mu \in \mathcal{M}$  reduces to “the universe is computable”: as weak an assumption as one can make. For technical details on Kolmogorov complexity, see [HQC24, §2.7].

The class of all computable measures is countable but not enumerable (by the halting problem, you can’t decide which Turing machines define total functions), so one can’t even formally write  $\sum_{\nu \in \mathcal{M}} \dots$ , as the decision problem of determining if  $\nu \stackrel{?}{\in} \mathcal{M}$  is undecidable. A faithful construction of  $\xi_U$  requires expanding  $\mathcal{M}$  to all *lower-semicomputable semimeasures* (Levin), at which point  $\xi_U$  becomes a semimeasure rather than a measure:  $\sum_x \xi_U(x) \leq 1$  instead of  $= 1$ , conditionals may not sum to 1, and every other proof in this sheet gains a side-case. This also gives the nice benefit that  $\xi \in \mathcal{M}$ .

We dodge all of this:  $\mathcal{M}$  is just *some* countable set of proper measures with  $\mu \in \mathcal{M}$ , and we assume  $K(\nu)$  is defined for each  $\nu \in \mathcal{M}$  when we need it. We never need  $\xi$  itself to be in  $\mathcal{M}$ , nor do we need its computability, so we don’t ask. See [Hut05, §2.4.3] for the full Levin construction.

## D. Best choice of $\hat{\mu} \in \mathcal{M}$ .

The bound is valid for every  $\hat{\mu} \in \mathcal{M}$ . A tempting question: which  $\hat{\mu}$  gives the tightest bound? The answer is genuinely setting-dependent.

*Finite horizon  $n$ .* The minimizer of the RHS is

$$\hat{\mu}_n := \arg \min_{\nu \in \mathcal{M}} \left\{ -\ln w_\nu + D_n(\mu \parallel \nu) \right\}.$$

This is well-defined (finitely many terms, attained when  $\mathcal{M}$  is finite), but the answer depends on  $n$ : for small  $n$  the prior term dominates and a high-prior coarse model wins; for large  $n$  the fit term dominates and the best per-step approximator wins. Both regimes are correct.

*Asymptotic rate.* A horizon-free analogue minimizes the per-step KL rate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t \leq n} d_t(\mu \parallel \nu).$$

This is the right notion when one cares about the linear-growth slope of the bound. But it has two drawbacks: the limit need not exist for non-stationary  $\mu$  (one can substitute  $\liminf$  but at the cost of clarity), and it discards the prior  $w_\nu$  entirely, so the minimizer is not the same as  $\hat{\mu}_n$  at any finite  $n$ , since only the slope is minimized.

*Stationary  $\mu$ .* The per-step KLs  $d_t(\mu \parallel \nu)$  are constant in  $t$ , so all three notions essentially agree:  $\hat{\mu} = \arg \min_{\nu \in \mathcal{M}} D_{\text{KL}}(\mu \parallel \nu)$ , where the KL is taken under the (common) one-step conditional law. This is the case where “ $\hat{\mu}$  = projection of  $\mu$  onto  $\mathcal{M}$  in KL” has unambiguous meaning.

*Upshot.* The bound holds for all  $\hat{\mu} \in \mathcal{M}$ , and one is free to take the infimum on the RHS over  $\hat{\mu}$ . Which  $\hat{\mu}$  actually attains that infimum is a separate question whose answer depends on horizon, prior, and any structural assumptions on  $\mu$ .

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